



Yash Soni

Roll No.: 2026PSP0019

M.Tech - Communication and Signal Processing

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EDUCATION

•Indian Institute of Technology Jammu

2026 – 2028

M.Tech in Communication and Signal Processing CGPA: -

Jammu, J & K

•Swami Keshvanand Institute of Technology (SKIT)

2020 – 2024

B.Tech in Electrical Engineering

CGPA: 9.57

Jaipur, Rajasthan

EXPERIENCE

•Indian Institute of Science (IISc)

June 2025 – July 2026

Junior Research Fellow

Bengaluru, Karnataka

- * Real-time Speech Denoising implementation on the 16 GB Windows equipped system and demonstrated the work to BEL officials.
- * Worked on the Speech Enhancement via radar signals which leads to a publication in a highly reputed academic conference ICASSP 2026.
- * Engineered a radar-based speech enhancement framework based on U-Net architectures with positional embeddings, and obtaining an aggregated score of 0.26.
- * Conducted the research on the road surface monitoring under low-visibility conditions and submerged conditions.
- * Executed research on low-power mmWave sensing signal analysis techniques, which ultimately improves the accuracy of intelligent sensing.

•CSIR - Central Electronics Engineering Research Institute (CSIR - CEERI)

August 2024 – June 2025

Project Associate

Pilani, Rajasthan

- * Implemented range analysis, range-doppler mapping, and range-angle mapping of the detected targets leveraging FMCW radar data with a range resolution of 4.5 cm.
- * Built deep learning models trained on a 4,000–5,000 sample dataset, attaining high multi-class accuracy across multiple radar target classes.
- * Demonstrated robust detection performance up to 30 m with minimal estimation error through external validation experiments.
- * Mentored and supervised 6 research interns, and 2 of them contributed to a journal-level manuscript submission.
- * Designed and deployed an interactive MATLAB GUI enabling post-data visualization of Range, Doppler, and Angle maps with adaptive CFAR tuning and a latency of 0.5 s for each frame.

PROJECTS

•Tariff Framework for EV Charging Station

Mar 2024 – May 2024

Graduation Project / Dr. Pooja Jain, Associate Professor, Dept. of EE, SKIT

- * Built and validated an hourly load forecasting model through machine learning algorithms on the American grid data and achieved 92% prediction accuracy.
- * Collaboratively conceptualised a strategic tariff framework to improve tariff structuring for EVCS, resulting in a 10% reduction in EV load and 5% increase in surcharge during peak-hours.

TECHNICAL SKILLS

- **Programming:** Python, C, MATLAB
- **Development Board:** AWR2944EVM, AWR1843BOOST, Raspberry Pi, Nicla Vision
- **Simulation Tools:** MATLAB GUI & MATLAB Simulink
- **Tools:** Google Teachable Machine, Google Colab, Canva, MS Office (Word, Excel, Powerpoint), LaTeX

KEY COURSES TAKEN

- Digital Signal Processing, Communication System, Signal Theory & RF Systems

Journal Publications

Yash Soni, Malhaar Goswami, Nishit Shetty, Dhiraj Sangwan "[Millimeter-wave Radar for Intelligent Sensing: A Comprehensive Review of Techniques, Applications, and Challenges](#)", *Computers and Electrical Engineering, Elsevier*

Conference Publications

Yash Soni, Alokendu Mazumder, Prasant Misra, Punit Rathore "[Posspec-Net: Enhancing radar-based speech signals via U-Nets and Positional Vectors](#)", *2026 51st IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*

Yash Soni, Dhiraj Sangwan "[mmWave Radar-Based Human Activity Recognition via Reinforcement Learning](#)", *2025 4th IEEE International Conference on Microwave, Antenna, and Propagation Conference (MAPCON)*

Yash Soni, Harshita Jangir, Pranabesh Maji, Dhiraj Sangwan "[Real-Time Target Classification using mmWave Radar through Statistical Analysis of 1D FFT Data](#)", *2025 6th IEEE India Council International Subsections Conference (INDISCON)*

Yash Soni, Harshita Jangir, Dhiraj Sangwan "Reinforcement Learning based Target Classification in mmWave Radar Systems", [Presented](#) *2025 16th IEEE International Conference on Computing, Communication and Networking Technologies (ICCCNT)*

Yash Soni, Dhiraj Sangwan "[Empirical Evaluation of Obstacle Range Estimation Techniques in Diverse Environments using mmWave Radar](#)", *2025 IEEE International Conference on Innovation in Computing and Engineering (ICE) (Best Paper Award)*

Pooja Jain, Ankush Tandon, Tushar Soni, **Yash Soni**, Vanshika Gupta, Vaidehi Mudgal "[Dynamic Tariff Optimization for EV Charging Stations using Reinforcement Learning](#)", *2024 3rd IEEE International Conference on Power Electronics, Intelligent Control and Energy Systems (ICPEICES)*

POSITIONS OF RESPONSIBILITY

- Student Representative**, Career Development Services, IIT Jammu *2026-Present*
 - Managed test administration for Online Assessments (OAs), resolving operational issues and ensuring process compliance.
 - Partnered with visiting hiring teams and campus leadership to deliver a streamlined, professional recruitment experience for candidates and recruiters.
- Student Coordinator**, IE(I) Student Chapter, SKIT Jaipur *2021 – 2024*
 - Spearheaded a technical team of 10 members to organize and execute 10+ college and departmental events in both virtual and in-person modes.
 - Managed end-to-end event operations, cross-functional logistics, and team coordination to ensure seamless execution throughout tenure.

ACHIEVEMENTS

- Awarded the Best Paper Award at an academic conference (ICE-2025)
- Two-time Semester Gold Medalist in Graduation.